Answers

1.**5**	2.**A**	3.**8**	4.**3**	5.**4**	6.**B**	7.**A**	8.**C**
9.**6**	10.**D**	11.**A**	12.**7**	13.**E**	14.**B**	15.**D**	16.**E**
17.**D**	18.**B**	19.**D**	20.**D**	21.**D**	22.**A**	23.**C**	24.**C**
25.**C**	26.**D**	27.**C**	28.**C**	29.**A**	30.**C**	31.**D**	32.**C**
33.**B**	34.**C**	35.**C**	36.**C**	37.**D**	38.**C**	39.**A**	40.**B**
41.**1**	42.**4**	43.**4**	44.**2**	45.**A**	46.**A**	47.**B**	48.**D**
49.**D**	50.**A**	51.**A**	52.**A**	53.**A**	54.**6**	55.**6**	56.**D**
57.**D**	58.**D**	59.**D**	60.**B**	61.**D**			

Explanations

1.**5**
We are told that there are six teams, that are divided into two groups.
Teams in the same group will play each other only once, and teams in different group will play each other twice.

Calculating the combinations, there is going to be $^3C_2$ games among teams in the same group among them, and since there is two groups, total such games will be 6.

Now, teams in different group play each other twice. Calculating the combinations for this,
From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is 3x3 which is 9. And since they play each other twice, that is 9+9 games of this combination.

Total number of games is 18+6=24
It is given that each team plays one game in each round, that means there is going to be 3 matchups in each round. And given, there is 24 games to played in this format, the number of rounds will be 24/3=8

The tournament will have 8 rounds.

Now that we know there are going to be 8 rounds in the tournament, let us identify the teams in a particular group, that will help us build the matchups.

We are told that Round 8 teams from different groups play each other, and Teams 1 and 5 play only once. This means, 1 and 5 have to be on the same group. It is also told that 4 and 6 play each other twice, that means 4 and 6 have to be in different groups. Looking at the matches from Round 8 that is given to us, 3 played 6 and 2 played 5. We already know 1 and 5 are in the same group, so 2 must be in the other group. Among 3 and 6, if we were to place 3 in the group with 1 and 5, 4 and 6 would have to be in the same group, which is not possible, hence 6 is with 1 and 5, giving us the final combination of groups.

Group-1	Group-2
1	2
5	3
6	4

Now, using the given information to build the matchups for the 8 rounds.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
Round-5	**Round-6**	**Round-7**	**Round-8**
	1 vs 6		3 vs 6
			2 vs 5

Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
	2 vs 3		
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6		3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

We are told that Round 4 and 7 are identical, that means they are the matchups between teams from two different groups,
We look at the matchups that are remaining among the 6 teams where both the games are left to play.

Right away we identify that 6 is yet to play 2 twice and 5 once. We are looking for teams playing twice, so both Round 4 and 7 has a matchups between 2 and 6. This means 6 will play 5 in Round 3 and using that we can identify the third matchup in Round 3 as well.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	2 vs 6
	1 vs 5	5 vs 6	
	2 vs 3	1 vs 2	
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	2 vs 6
1 vs 2	1 vs 5	5 vs 6	1 vs 3
3 vs 5	2 vs 3	1 vs 2	4 vs 5
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5	2 vs 4	1 vs 3	2 vs 5
1 vs 4	3 vs 5	4 vs 5	1 vs 4

. This is the final table.

Team that played Team 6 in Round 3 was Team 5.

VIDEO SOLUTION

2. **A**

We are told that there are six teams, that are divided into two groups.
Teams in the same group will play each other only once, and teams in different group will play each other twice.

Calculating the combinations, there is going to be $^3C_2$ games among teams in the same group among them, and since there is two groups, total such games will be 6.

Now, teams in different group play each other twice. Calculating the combinations for this,
From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is 3x3 which is 9. And since they play each other twice, that is 9+9 games of this combination.

Total number of games is 18+6=24
It is given that each team plays one game in each round, that means there is going to be 3 matchups in each round. And given, there is 24 games to played in this format, the number of rounds will be 24/3=8

The tournament will have 8 rounds.

Now that we know there are going to be 8 rounds in the tournament, let us identify the teams in a particular group, that will help us build the matchups.

We are told that Round 8 teams from different groups play each other, and Teams 1 and 5 play only once. This means, 1 and 5 have to be on the same group. It is also told that 4 and 6 play each other twice, that means 4 and 6 have to be in different groups. Looking at the matches from Round 8 that is given to us, 3 played 6 and 2 played 5. We already know 1 and 5 are in the same group, so 2 must be in the other group. Among 3 and 6, if we were to place 3 in the group with 1 and 5, 4 and 6 would have to be in the same group, which is not possible, hence 6 is with 1 and 5, giving us the final combination of groups.

Group-1	Group-2
1	2
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6	4

Now, using the given information to build the matchups for the 8 rounds.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
Round-5	**Round-6**	**Round-7**	**Round-8**
	1 vs 6		3 vs 6
			2 vs 5

Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

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4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
	2 vs 3		
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6		3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

We are told that Round 4 and 7 are identical, that means they are the matchups between teams from two different groups,

We look at the matchups that are remaining among the 6 teams where both the games are left to play.

Right away we identify that 6 is yet to play 2 twice and 5 once. We are looking for teams playing twice, so both Round 4 and 7 has a matchups between 2 and 6. This means 6 will play 5 in Round 3 and using that we can identify the third matchup in Round 3 as well.

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4 vs 6	4 vs 6	3 vs 4	2 vs 6
	1 vs 5	5 vs 6	
	2 vs 3	1 vs 2	
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	2 vs 6
1 vs 2	1 vs 5	5 vs 6	1 vs 3
3 vs 5	2 vs 3	1 vs 2	4 vs 5
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5	2 vs 4	1 vs 3	2 vs 5
1 vs 4	3 vs 5	4 vs 5	1 vs 4

. This is the final table.

2, 3 and 4 were part of the same group. 5 is the answer.

VIDEO SOLUTION

3. **8**

We are told that there are six teams, that are divided into two groups.
Teams in the same group will play each other only once, and teams in different group will play each other twice.

Calculating the combinations, there is going to be $^3C_2$ games among teams in the same group among them, and since there is two groups, total such games will be 6.

Now, teams in different group play each other twice. Calculating the combinations for this,
From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is 3x3 which is 9. And since they play each other twice, that is 9+9 games of this combination.

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It is given that each team plays one game in each round, that means there is going to be 3 matchups in each round. And given, there is 24 games to played in this format, the number of rounds will be 24/3=8

The tournament will have 8 rounds.

Now that we know there are going to be 8 rounds in the tournament, let us identify the teams in a particular group, that will help us build the matchups.

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Group-1	Group-2
1	2
5	3
6	4

Now, using the given information to build the matchups for the 8 rounds.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
Round-5	**Round-6**	**Round-7**	**Round-8**
	1 vs 6		3 vs 6
			2 vs 5

Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

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4 vs 6	4 vs 6	3 vs 4	2 vs 6
	1 vs 5	5 vs 6	
	2 vs 3	1 vs 2	
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	2 vs 6
1 vs 2	1 vs 5	5 vs 6	1 vs 3
3 vs 5	2 vs 3	1 vs 2	4 vs 5
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5	2 vs 4	1 vs 3	2 vs 5
1 vs 4	3 vs 5	4 vs 5	1 vs 4

. This is the final table.

The tournament will have 8 rounds.

VIDEO SOLUTION

4. **3**
We are told that there are six teams, that are divided into two groups.
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Group-1	Group-2
1	2
5	3
6	4

Now, using the given information to build the matchups for the 8 rounds.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
Round-5	**Round-6**	**Round-7**	**Round-8**
	1 vs 6		3 vs 6
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4 vs 6	4 vs 6	3 vs 4	2 vs 6
	1 vs 5	5 vs 6	
	2 vs 3	1 vs 2	
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.

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4 vs 6	4 vs 6	3 vs 4	2 vs 6
1 vs 2	1 vs 5	5 vs 6	1 vs 3
3 vs 5	2 vs 3	1 vs 2	4 vs 5
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5	2 vs 4	1 vs 3	2 vs 5
1 vs 4	3 vs 5	4 vs 5	1 vs 4

. This is the final table.

The team that played Team 1 in Round 7 is 3.

VIDEO SOLUTION

5. **4**

We are told that there are six teams, that are divided into two groups.
Teams in the same group will play each other only once, and teams in different group will play each other twice.

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Now, teams in different group play each other twice. Calculating the combinations for this,
From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is 3x3 which is 9. And since they play each other twice, that is 9+9 games of this combination.

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Group-1	Group-2
1	2
5	3
6	4

Now, using the given information to build the matchups for the 8 rounds.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
Round-5	**Round-6**	**Round-7**	**Round-8**
	1 vs 6		3 vs 6
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Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
	2 vs 3		
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6		3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

We are told that Round 4 and 7 are identical, that means they are the matchups between teams from two different groups,
We look at the matchups that are remaining among the 6 teams where both the games are left to play.

Right away we identify that 6 is yet to play 2 twice and 5 once. We are looking for teams playing twice, so both Round 4 and 7 has a matchups between 2 and 6. This means 6 will play 5 in Round 3 and using that we can identify the third matchup in Round 3 as well.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	2 vs 6
	1 vs 5	5 vs 6	
	2 vs 3	1 vs 2	
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	2 vs 6
1 vs 2	1 vs 5	5 vs 6	1 vs 3
3 vs 5	2 vs 3	1 vs 2	4 vs 5
Round-5	**Round-6**	**Round-7**	**Round-8**
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5	2 vs 4	1 vs 3	2 vs 5
1 vs 4	3 vs 5	4 vs 5	1 vs 4

. This is the final table.

The team that played Team 1 in Round 5 is 4.

VIDEO SOLUTION

6.**B**
After second move i will be having from a quarter cell of board to make a move.

Since left quarters have cells with values 1, Hence i will choose to retain right in my first move so that second person can force my profit to be either 2 or 3 not 1.

So now after my first move of retaining right, we have two quarters

Now intelligently he will make a move to retain with lower half so that at fourth move i can retain with choices of 2 and 3 not 2 and 4.

VIDEO SOLUTION

7.**A**
After retaining right in first move, second person will have 2 quarters to choose in which minimum value is 2 and 3.
If he retains upper half, i will retain right as to maximize my profit of either 4 or 7.
If he retains lower half, i will retain left as to maximize my profit of either 3 or 8.
Hence whatever he retains, i won't have a profit less than 3.

VIDEO SOLUTION

8.**C**
After my first move of retaining right board will be like

2	4
6	7
3	2
8	4

Now second person will force me to retain with the lowest value possible i.e. 2 or 3
So if he retains lower half and i retain with left one (as mentioned) then he can only force me to pick 3.
Or if he retains upper half and i retain with left one(as mentioned) then he can retain upper half so i will be

having minimized profit of 2.
Hence second person will have moves as "Retain (upper half) and retain (upper half)"

VIDEO SOLUTION

9. **6**
We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

6 + Badal + Clive + 11 + 9 = 40

Badal + Clive = 14

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be 1 + 1 + 2 + 2 = 6. It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

Badal + 1 + 1 + Ehsan = 6

Badal + Ehsan = 4

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

Alia + 2 + Clive + Ehsan = 11

Alia + Clive + Ehsan = 9

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C						8
D	3	2	3		3	11
E						9
TAPS RECEIVED	9	a		a	a	40

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as 2 + 2 + 1 + 3 = 8. We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

9 + 8 + c + 8 + 8 = 40

c = 7

So, the number of taps received by Clive is 7.

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C		1/3				8
D	3	2	3		3	11
E		3/1				9
TAPS RECEIVED	9	8	7	8	8	40

We are given that the answer by Alia does not match the answers people gave to her question. We know that the taps by Badal have to be 1, 1, 2 and 2 in some order. We know that Alia tapped twice to Badal's question, so Badal cannot tap twice to Alia's question, which means Badal tapped once for Alia's question. Now the sum of Clive and Ehsan's taps for Alia's question has to be 9 - 3 - 1 = 5. So, they have to be 2 and 3 in some order. We also know that Alia's answer to Ehsan's question is 2, so Ehsan has only one option to answer Alia, which is 3, and Clive's answer to Alia's question is 2.

Badal and Ehsan's answer to Clive's question has to be 1 and 2 in some order. Similarly, Badal and Clive's answers to Ehsan's question have to be 1 and 3 in some order.

We can also calculate the sum of taps by Badal, Clive and Ehsan to Dilshan's question is 8 - 1 = 7. So, their taps have to be 2, 2 and 3 in some order as this is the only possibility satisfying the condition of at least one Yes, one No and one Maybe for every question. We know that a tap Badal has to be either one or two, so the only possibility for Badal's answer to Dilshan's question is 2 and the other two has to be two and three in some order.

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B	1		2/1	2	1/2	6
C	2	1/3		2/3	2/1	8
D	3	2	3		3	11
E	3	3/1	1/2	3/2		9
TAPS RECEIVED	9	8	7	8	8	40

We are also given that the answer by Ehsan does not match the answers people gave to his question. Dilshan's answer to Ehsan's question is 3, so the answer by Ehsan to Dilshan's question has to be 2.

The answer by Clive to Badal's question has to be 1, as if it is 3, then the answer to Ehsan's question by Clive becomes 0, which is not possible. With that value, we can determine all the other values and putting the values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B	1		2	2	1	6
C	2	1		3	2	8
D	3	2	3		3	11
E	3	3	1	2		9
TAPS RECEIVED	9	8	7	8	8	40

The total number of Yes responses is equal to the number of 1's in the table, which is 6.

Hence, the correct answer is 6.

VIEW SOLUTION

10. **D**

We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

6 + Badal + Clive + 11 + 9 = 40

Badal + Clive = 14

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be 1 + 1 + 2 + 2 = 6. It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

Badal + 1 + 1 + Ehsan = 6

Badal + Ehsan = 4

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

Alia + 2 + Clive + Ehsan = 11

Alia + Clive + Ehsan = 9

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C						8
D	3	2	3		3	11
E						9
TAPS RECEIVED	9	a		a	a	40

Alia and Badal are the only people with an equal number of taps.

Hence, the correct answer is option D.

VIEW SOLUTION

11. **A**

We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

6 + Badal + Clive + 11 + 9 = 40

Badal + Clive = 14

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be 1 + 1 + 2 + 2 = 6. It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

Badal + 1 + 1 + Ehsan = 6

Badal + Ehsan = 4

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

Alia + 2 + Clive + Ehsan = 11

Alia + Clive + Ehsan = 9

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C						8
D	3	2	3		3	11
E						9
TAPS RECEIVED	9	a		a	a	40

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as 2 + 2 + 1 + 3 = 8. We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

9 + 8 + c + 8 + 8 = 40

c = 7

So, the number of taps received by Clive is 7.

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C		1/3				8
D	3	2	3		3	11
E		3/1				9
TAPS RECEIVED	9	8	7	8	8	40

We are given that the answer by Alia does not match the answers people gave to her question. We know that the taps by Badal have to be 1, 1, 2 and 2 in some order. We know that Alia tapped twice to Badal's question, so Badal cannot tap twice to Alia's question, which means Badal tapped once for Alia's question. Now the sum of Clive and Ehsan's taps for Alia's question has to be 9 - 3 - 1 = 5. So, they have to be 2 and 3 in some order. We also know that Alia's answer to Ehsan's question is 2, so Ehsan has only one option to answer Alia, which is 3, and Clive's answer to Alia's question is 2.

Badal and Ehsan's answer to Clive's question has to be 1 and 2 in some order. Similarly, Badal and Clive's answers to Ehsan's question have to be 1 and 3 in some order.

We can also calculate the sum of taps by Badal, Clive and Ehsan to Dilshan's question is 8 - 1 = 7. So, their taps have to be 2, 2 and 3 in some order as this is the only possibility satisfying the condition of at least one Yes, one No and one Maybe for every question. We know that a tap Badal has to be either one or two, so the only possibility for Badal's answer to Dilshan's question is 2 and the other two has to be two and three in some order.

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B	1		2/1	2	1/2	6
C	2	1/3		2/3	2/1	8
D	3	2	3		3	11
E	3	3/1	1/2	3/2		9
TAPS RECEIVED	9	8	7	8	8	40

We are also given that the answer by Ehsan does not match the answers people gave to his question. Dilshan's answer to Ehsan's question is 3, so the answer by Ehsan to Dilshan's question has to be 2.

The answer by Clive to Badal's question has to be 1, as if it is 3, then the answer to Ehsan's question by Clive becomes 0, which is not possible. With that value, we can determine all the other values and putting the values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B	1		2	2	1	6
C	2	1		3	2	8
D	3	2	3		3	11
E	3	3	1	2		9
TAPS RECEIVED	9	8	7	8	8	40

Clive's answer to Ehsan's question is No, which is denoted by 2.

Hence, the correct answer is option A.

VIEW SOLUTION

12. **7**

We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

6 + Badal + Clive + 11 + 9 = 40

Badal + Clive = 14

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be 1 + 1 + 2 + 2 = 6. It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

Badal + 1 + 1 + Ehsan = 6

Badal + Ehsan = 4

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

Alia + 2 + Clive + Ehsan = 11

Alia + Clive + Ehsan = 9

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C						8
D	3	2	3		3	11
E						9
TAPS RECEIVED	9	a		a	a	40

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as 2 + 2 + 1 + 3 = 8. We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

9 + 8 + c + 8 + 8 = 40

c = 7

So, the number of taps received by Clive is 7.

Hence, the correct answer is 7.

VIEW SOLUTION

13. **E**

There are a total of ${}^6C_2$ matches => 15 matches. The first 9 matches are held in the first stage and remaining 6 in the second stage.

From the information given, we can conclude that the following matches were held in first stage:

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won)

One team won all matches. As B, C, D E and F have lost at least one match each, A won all three matches. As A, B, D, E have won at least one match, C and F lost both matches.

From the matches already deduced, we can see that A needs to play 2 more matches, B two more matches and C and F one match each. As C and F lose all matches in stage 1, they cannot play against each other. F did not play against the leader i.e. A. Hence, the remaining matches are A-B (A won), A-C (A won), B-F (B won).

Thus, the stage 1 matches are

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won), A-B (A won), A-C (A won), B-F (B won)

Thus Stage 2 matches are D-B, D-E, E-A, F-A, B-C and C-F (all matches - stage 1 matches)

As A lost both matches, F and E must have won the match vs A. As F won against A, F won both its matches and C lost both its matches. One more team lost both its matches. As B, E and F have won at least one match and A and C have been discussed previously, D must have lost both matches. Hence, stage 2 results are:

Stage 2: D-B (B won), D-E (E won), E-A (E won), F-A (F won), B-C (B won) and C-F (F won)

Hence, the wins by each team are A(3), B(4), C(0), D(2), E(4), F(2). Hence, most wins are by B and E.

VIDEO SOLUTION

14. **B**

There are a total of ${}^6C_2$ matches => 15 matches. The first 9 matches are held in the first stage and remaining 6 in the second stage.

From the information given, we can conclude that the following matches were held in first stage:

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won)

One team won all matches. As B, C, D E and F have lost at least one match each, A won all three matches. As A, B, D, E have won at least one match, C and F lost both matches.

From the matches already deduced, we can see that A needs to play 2 more matches, B two more matches and C and F one match each. As C and F lose all matches in stage 1, they cannot play against each other. F did not play against the leader i.e. A. Hence, the remaining matches are A-B (A won), A-C (A won), B-F (B won).

Thus, the stage 1 matches are

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won), A-B (A won), A-C (A won), B-F (B won)

Thus Stage 2 matches are D-B, D-E, E-A, F-A, B-C and C-F (all matches - stage 1 matches)

As A lost both matches, F and E must have won the match vs A. As F won against A, F won both its matches and C lost both its matches. One more team lost both its matches. As B, E and F have won at least one match and A and C have been discussed previously, D must have lost both matches. Hence, stage 2 results are:

Stage 2: D-B (B won), D-E (E won), E-A (E won), F-A (F won), B-C (B won) and C-F (F won)

Hence, the two teams that won against stage 1 leader A are E and F.

VIDEO SOLUTION

15. **D**

There are a total of ${}^6C_2$ matches => 15 matches. The first 9 matches are held in the first stage and remaining 6 in the second stage.

From the information given, we can conclude that the following matches were held in first stage:

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won)

One team won all matches. As B, C, D E and F have lost at least one match each, A won all three matches. As A, B, D, E have won at least one match, C and F lost both matches.

From the matches already deduced, we can see that A needs to play 2 more matches, B two more matches and C and F one match each. As C and F lose all matches in stage 1, they cannot play against each other. F did not play against the leader i.e. A. Hence, the remaining matches are A-B (A won), A-C (A won), B-F (B won).

Thus, the stage 1 matches are

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won), A-B (A won), A-C (A won), B-F (B won)

Thus Stage 2 matches are D-B, D-E, E-A, F-A, B-C and C-F (all matches - stage 1 matches)

As A lost both matches, F and E must have won the match vs A. As F won against A, F won both its matches and C lost both its matches. One more team lost both its matches. As B, E and F have won at least one match and A and C have been discussed previously, D must have lost both matches. Hence, stage 2 results are:

Stage 2: D-B (B won), D-E (E won), E-A (E won), F-A (F won), B-C (B won) and C-F (F won)

Hence, the teams that won both of their stage 2 matches are B, E and F.

VIDEO SOLUTION

16. **E**

There are a total of ${}^6C_2$ matches => 15 matches. The first 9 matches are held in the first stage and remaining 6 in the second stage.

From the information given, we can conclude that the following matches were held in first stage:

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won)

One team won all matches. As B, C, D E and F have lost at least one match each, A won all three matches. As A, B, D, E have won at least one match, C and F lost both matches.

From the matches already deduced, we can see that A needs to play 2 more matches, B two more matches and C and F one match each. As C and F lose all matches in stage 1, they cannot play against each other. F did not play against the leader i.e. A. Hence, the remaining matches are A-B (A won), A-C (A won), B-F (B won).

Thus, the stage 1 matches are

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won), A-B (A won), A-C (A won), B-F (B won)

Thus Stage 2 matches are D-B, D-E, E-A, F-A, B-C and C-F (all matches - stage 1 matches)

As A lost both matches, F and E must have won the match vs A. As F won against A, F won both its matches and C lost both its matches. One more team lost both its matches. As B, E and F have won at least one match and A and C have been discussed previously, D must have lost both matches. Hence, stage 2 results are:

Stage 2: D-B (B won), D-E (E won), E-A (E won), F-A (F won), B-C (B won) and C-F (F won)

Hence, the wins by each team are A (3), B(4), C(0), D(2), E(4), F(2). Hence, D and F won exactly 2 matches.

VIDEO SOLUTION

17. **D**

Germany and Argentina have 2 games each and Spain and Pakistan have won 1 game each. We also know that Germany beat Spain.

So Germany won its second game against one among SA/ NZ.

Similarly Argentina must have won one game against Pakistan and another against SA/ NZ.

Argentina scored 2 goals and conceded 0 goals. So it must have won both the games 1-0.

NZ conceded 6 goals. The only possibility is it lost 5-1 to Spain and lost 1-0 to Argentina.

As Spain conceded 2 goals it must have lost its opening game to Germany by 1-0.

Now, the results of all the games can be known.

Following are the possible match result after 2 rounds :

GER	1-0	SPA
GER	2-1	SA
ARG	1-0	PAK
ARG	1-0	NZ
SPA	5-1	NZ
PAK	2-0	SA

HENCE OPTION D IS THE CORRECT ANSWER.

VIEW SOLUTION

18. **B**

Following are the possible match result of 2 rounds :

GER	1-0	SPA
GER	2-1	SA
ARG	1-0	PAK
ARG	1-0	NZ
SPA	5-1	NZ
PAK	2-0	SA

Hence option D is the correct answer.

VIEW SOLUTION

19. **D**
Data is inconsistent as from the statements we can infer that Spain, Germany, Argentina and Pakistan won thei 5th round matches. But only 3 teams can win any round.Hence the following question can't be answered.

VIEW SOLUTION

20. **D**
Data is inconsistent as from the statements we can infer that Spain, Germany, Argentina and Pakistan won their 5th round matches. But only 3 teams can win any round as there are total of 6 teams.Hence the following question can't be answered.

VIEW SOLUTION

21. **D**
There are no upsets in 1st round, so top 16 players go to the 2nd round.

In second round, nth player would play (17-n)th player. Now, match Nos. 6, 7, and 8 of the second round result in upsets.

So the people in the quarter finals would be 1,2,3,4,5,11,10,9.

So in quarterfinal 2nd seeeded player would play (9-2) = 7th position player.

Here no. 10 is in the 7th position and No. 10 player is venus williams.

Hence option D.

VIDEO SOLUTION

22. **A**
According to given condition, players of following ranking will go to the 2nd round 1,31,3,29,5,27,7,25,9,23,11,21,13,19,15,17.

Out of these players going to 3rd round / quarter finals are 1,15,3,13,5,11,7,9.

If maria (no.1) goes to semifinal she'll face any on eout of the 13/5 th no. player in semis.

13th seed player is the lowest so Anastasia Myskina.

VIDEO SOLUTION

23. **C**
If the top eight seeds make it to the quarterfinals then one out of Kim Clijsters or svetana kuznetsova will compete with maria sharapoava in semis.

So now if maria reaches semis, she'll beat any one out of the two and both will definitely not play against Maria Sharapova in the final, if she reaches .

VIDEO SOLUTION

24. **C**

If Elena Dementieva and Serena Williams lose in the second round, Nadia petrova and patty schnyder will go through.

Hence ,in the quarter finals, following seed no. players will be in quarter finals: 1,2,3,4,5,11,7,9.

So, now Maria Sharapova is ranked is 1 so she'll play 9th seed player who is Nadia Petrova.

Hence option C.

VIDEO SOLUTION

25. **C**

Let us arrange the players in the order in which they throw in each round.

Round 1: Here the players throw in order of their initial seeds so the order is as follows:

Throw	Player
1	P1
2	P2
3	P3
4	P4
5	P5
6	P6
7	P7
8	P8
9	P9
10	P10

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

Round 2: P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

Throw	Round 1	Round 2
1	P1	P7
2	P2	P5
3	P3	P9
4	P4	P1
5	P5	P6
6	P6	P3
7	P7	P2
8	P8	P4
9	P9	P8
10	P10	P10

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

Round 3: In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position. P1 > P7 > P5 > P9.

So at the end of round 3, the ranks are as follows:

P1
P7
P5
P9
P8/P10
P6
P3
P2
P4

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 - > 5 AND 5 - > 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

Rank after Round 3	
Case 1	Case 2
P10	P1
P1	P10
P7	P7
P5	P5
P9	P9
P8	P8

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by x and reach the top 3 positions. However, the top 3 positions' distances are in AP.

P1 - 88.6 + x

P7 - 87.2 + x

P5 - 86.4 + x

P9 - 84.1 + x

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

P1 - 88.6

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

89.6 - 87.2 = 2.4

87.6 - 86.4 = 1.2

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

Hence, P8 and P10 got the doubles.

VIDEO SOLUTION

26. **D**
Let us arrange the players in the order in which they throw in each round.

Round 1: Here the players throw in order of their initial seeds so the order is as follows:

Throw	Player
1	P1
2	P2
3	P3
4	P4
5	P5
6	P6
7	P7
8	P8
9	P9
10	P10

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

Round 2: P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

Throw	Round 1	Round 2
1	P1	P7
2	P2	P5
3	P3	P9
4	P4	P1
5	P5	P6
6	P6	P3
7	P7	P2
8	P8	P4
9	P9	P8
10	P10	P10

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

Round 3: In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position. P1 > P7 > P5 > P9.

So at the end of round 3, the ranks are as follows:

P1
P7
P5
P9
P8/P10
P6
P3
P2
P4

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 - > 5 AND 5 - > 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

Rank after Round 3	
Case 1	Case 2
P10	P1
P1	P10
P7	P7
P5	P5
P9	P9
P8	P8

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by x and reach the top 3 positions. However, the top 3 positions' distances are in AP.

P1 - 88.6 + x

P7 - 87.2 + x

P5 - 86.4 + x

P9 - 84.1 + x

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

P1 - 88.6

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

89.6 - 87.2 = 2.4

87.6 - 86.4 = 1.2

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

Hence, P1 won the silver medal.

VIDEO SOLUTION

27. **C**
Let us arrange the players in the order in which they throw in each round.

Round 1: Here the players throw in order of their initial seeds so the order is as follows:

Throw	Player
1	P1
2	P2
3	P3
4	P4
5	P5
6	P6
7	P7
8	P8
9	P9
10	P10

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

Round 2: P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

Throw	Round 1	Round 2
1	P1	P7
2	P2	P5
3	P3	P9
4	P4	P1
5	P5	P6
6	P6	P3
7	P7	P2
8	P8	P4
9	P9	P8
10	P10	P10

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

Round 3: In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position. P1 > P7 > P5 > P9.

So at the end of round 3, the ranks are as follows:

P1
P7
P5
P9
P8/P10
P6
P3
P2
P4

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 - > 5 AND 5 - > 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

Rank after Round 3	
Case 1	Case 2
P10	P1
P1	P10
P7	P7
P5	P5
P9	P9
P8	P8

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by x and reach the top 3 positions. However, the top 3 positions' distances are in AP.

P1 - 88.6 + x

P7 - 87.2 + x

P5 - 86.4 + x

P9 - 84.1 + x

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

P1 - 88.6

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

89.6 - 87.2 = 2.4

87.6 - 86.4 = 1.2

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

P7 improved by a total of 2.4.

VIDEO SOLUTION

28. **C**

Let us arrange the players in the order in which they throw in each round.

Round 1: Here the players throw in order of their initial seeds so the order is as follows:

Throw	Player
1	P1
2	P2
3	P3
4	P4
5	P5
6	P6
7	P7
8	P8
9	P9
10	P10

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

Round 2: P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

Throw	Round 1	Round 2
1	P1	P7
2	P2	P5
3	P3	P9
4	P4	P1
5	P5	P6
6	P6	P3
7	P7	P2
8	P8	P4
9	P9	P8
10	P10	P10

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

Round 3: In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position. P1 > P7 > P5 > P9.

So at the end of round 3, the ranks are as follows:

P1
P7
P5
P9
P8/P10
P6
P3
P2
P4

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 - > 5 AND 5 - > 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

Rank after Round 3	
Case 1	Case 2
P10	P1
P1	P10
P7	P7
P5	P5
P9	P9
P8	P8

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by x and reach the top 3 positions. However, the top 3 positions' distances are in AP.

P1 - 88.6 + x

P7 - 87.2 + x

P5 - 86.4 + x

P9 - 84.1 + x

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

P1 - 88.6

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

89.6 - 87.2 = 2.4

87.6 - 86.4 = 1.2

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

P8 comes between P9 and P6.

Hence, 82. 5 < P8 < 84.1

P8 = 82.7

VIDEO SOLUTION

29. **A**

Let us arrange the players in the order in which they throw in each round.

Round 1: Here the players throw in order of their initial seeds so the order is as follows:

Throw	Player
1	P1
2	P2
3	P3
4	P4
5	P5
6	P6
7	P7
8	P8
9	P9
10	P10

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

Round 2: P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

Throw	Round 1	Round 2
1	P1	P7
2	P2	P5
3	P3	P9
4	P4	P1
5	P5	P6
6	P6	P3
7	P7	P2
8	P8	P4
9	P9	P8
10	P10	P10

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

Round 3: In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position. P1 > P7 > P5 > P9.

So at the end of round 3, the ranks are as follows:

P1
P7
P5
P9
P8/P10
P6
P3
P2
P4

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 - > 5 AND 5 - > 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

Rank after Round 3	
Case 1	Case 2
P10	P1
P1	P10
P7	P7
P5	P5
P9	P9
P8	P8

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by x and reach the top 3 positions. However, the top 3 positions' distances are in AP.

P1 - 88.6 + x

P7 - 87.2 + x

P5 - 86.4 + x

P9 - 84.1 + x

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

P1 - 88.6

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

89.6 - 87.2 = 2.4

87.6 - 86.4 = 1.2

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

P7 the gold winner had already received rank 1 at the end of Round 5. Hence, he was the last one to throw in the tournament.

VIDEO SOLUTION

30. **C**

Let us arrange the players in the order in which they throw in each round.

Round 1: Here the players throw in order of their initial seeds so the order is as follows:

Throw	Player
1	P1
2	P2
3	P3
4	P4
5	P5
6	P6
7	P7
8	P8
9	P9
10	P10

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

Round 2: P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

Throw	Round 1	Round 2
1	P1	P7
2	P2	P5
3	P3	P9
4	P4	P1
5	P5	P6
6	P6	P3
7	P7	P2
8	P8	P4
9	P9	P8
10	P10	P10

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

Round 3: In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position. P1 > P7 > P5 > P9.

So at the end of round 3, the ranks are as follows:

P1
P7
P5
P9
P8/P10
P6
P3
P2
P4

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 - > 5 AND 5 - > 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

Rank after Round 3	
Case 1	Case 2
P10	P1
P1	P10
P7	P7
P5	P5
P9	P9
P8	P8

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by x and reach the top 3 positions. However, the top 3 positions' distances are in AP.

P1 - 88.6 + x

P7 - 87.2 + x

P5 - 86.4 + x

P9 - 84.1 + x

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

P1 - 88.6

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

89.6 - 87.2 = 2.4

87.6 - 86.4 = 1.2

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

P1 - 88.6 m.

VIDEO SOLUTION

31. **D**

There are 28 matches in the 1st round of 8 player group so in total 28 wins.To find this, we need the top 5 teams to win nearly equal matches. So Let each of the top 5 teams win 5 matches each and the remaining 3 matches are won by the bottom 2 teams. The qualifiers will be decided based on the tie breaking rules. Hence, even with 5 wins a team may end up not qualifying for the next round. Thus, option D is the correct answer.

VIDEO SOLUTION

32. **C**

In 1st stage there were 2 groups each group played 28 matches as ${}^{8}C_{2}$ = 28. So total 56 matches after 1st stage . IN 1st round of stage 2, 4 matches are played and in the next round we have 4 teams .The winner is one who defeats every team out of remaining 3 matches . so 3 matches more needed. In total 56+4+3=63.

VIDEO SOLUTION

33. **B**

Let us first try to find the maximum number of points with which a team cannot advance to the second stage.

For this to happen the top five teams should have highest possible equal number of points and one team does not advance after the tie breaker.

To maximize the points of top 5 teams all of the top 5 teams should win their matches with the bottom 3 teams. Now each team has 3 points.

The top 5 teams play 5C2 = 10 matches among themselves. There are 10 points to be won from these 10 matches.

Now as all the top 5 teams should have equal number of points each team gets 2 points from these games.

In all each top 5 team gets 3+2 = 5 points each.

In this case after getting 5 points also one of the top 5 teams will not advance to the second round.

Thus the maximum number of points with which a team cannot advance to the second stage is 5.

Therefore at least 6 wins are required for a team to guarantee its advancement to the next stage.

VIDEO SOLUTION

34. **C**

2nd stage 1st round has 8 teams , 2nd round will have 4 teams and 3rd round 2 teams where the winner wins .

VIDEO SOLUTION

35. **C**

Consider the below case.

Group 1			Group 2	
A	7		P	7
B	6		Q	6
C	5		R	5
D	2		S	4
E	2		T	3
F	2		U	2
G	2		V	1
H	2		W	0

If D wins the tournament he would have 5 wins. But if P loses in second round, he would have 7 wins. Hence option A is incorrect.

We can see that T has more wins than D. Hence option B is incorrect.

Consider the below scenario:

QF	Wins	SF	Wins	F	Wins	Total wins
A	1	A	1	A	1	3
B	1	B	1	B	0	2
C	1	C	0			1
D	1	D	0			1
P	0					0
Q	0					0
R	0					0
S	0					0

We can see that 2 teams have one win each in the second stage. Hence option D is incorrect.

Consider the following scenario.

Group 1			Group 2	
A	7		P	5
B	6		Q	5
C	5		R	5
D	2		S	5
E	2		T	5
F	2		U	1
G	2		V	1
H	2		W	1

If D wins the tournament he would have 5 wins. But T would lose the tournament with 5 wins. Hence option C is correct.

VIEW SOLUTION

36. **C**

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

Matches	Goals Scored	Players
Match-1		Bimal-1
Match-2	1	
Match-3		
Match-4	1	Harita-1
Match-5		Bimal-1
Match-6	1	Bimal-1
Match-7		Bimal-1
Match-8	1	Harita-1

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be a each and no. of goals scored in 1st and 5th match be b and c respectively.

Therefore, 2a+b+c= 8

If a=1, then b+c=6 therefore possible solutions for b and c will be 2 and 4 only

If a=2, then b+c=4 therefore possible solution for b and c will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is a=1, b=4 and c=2

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	
Match-3	1	
Match-4	1	Harita-1
Match-5	2	Bimal-1,
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	Amla/Sarita-1
Match-3	1	Amla/Sarita-1
Match-4	1	Harita-1
Match-5	2	Bimal-1, Amla/Sarita-1
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

VIDEO SOLUTION

37. **D**

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

Matches	Goals Scored	Players
Match-1		Bimal-1
Match-2	1	
Match-3		
Match-4	1	Harita-1
Match-5		Bimal-1
Match-6	1	Bimal-1
Match-7		Bimal-1
Match-8	1	Harita-1

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be a each and no. of goals scored in 1st and 5th match be b and c respectively.

Therefore, 2a+b+c= 8

If a=1, then b+c=6 therefore possible solutions for b and c will be 2 and 4 only

If a=2, then b+c=4 therefore possible solution for b and c will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is a=1, b=4 and c=2

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	
Match-3	1	
Match-4	1	Harita-1
Match-5	2	Bimal-1,
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	Amla/Sarita-1
Match-3	1	Amla/Sarita-1
Match-4	1	Harita-1
Match-5	2	Bimal-1, Amla/Sarita-1
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

VIDEO SOLUTION

38. **C**

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

Matches	Goals Scored	Players
Match-1		Bimal-1
Match-2	1	
Match-3		
Match-4	1	Harita-1
Match-5		Bimal-1
Match-6	1	Bimal-1
Match-7		Bimal-1
Match-8	1	Harita-1

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be a each and no. of goals scored in 1st and 5th match be b and c respectively.

Therefore, 2a+b+c= 8

If a=1, then b+c=6 therefore possible solutions for b and c will be 2 and 4 only

If a=2, then b+c=4 therefore possible solution for b and c will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is a=1, b=4 and c=2

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	
Match-3	1	
Match-4	1	Harita-1
Match-5	2	Bimal-1,
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	Amla/Sarita-1
Match-3	1	Amla/Sarita-1
Match-4	1	Harita-1
Match-5	2	Bimal-1, Amla/Sarita-1
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

VIDEO SOLUTION

39. **A**

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

Matches	Goals Scored	Players
Match-1		Bimal-1
Match-2	1	
Match-3		
Match-4	1	Harita-1
Match-5		Bimal-1
Match-6	1	Bimal-1
Match-7		Bimal-1
Match-8	1	Harita-1

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be a each and no. of goals scored in 1st and 5th match be b and c respectively.

Therefore, $2a+b+c= 8$

If $a=1$, then $b+c=6$ therefore possible solutions for b and c will be 2 and 4 only

If $a=2$, then $b+c=4$ therefore possible solution for b and c will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is $a=1$, $b=4$ and $c=2$

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	
Match-3	1	
Match-4	1	Harita-1
Match-5	2	Bimal-1,
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	Amla/Sarita-1
Match-3	1	Amla/Sarita-1
Match-4	1	Harita-1
Match-5	2	Bimal-1, Amla/Sarita-1
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

VIDEO SOLUTION

40. **B**

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

Matches	Goals Scored	Players
Match-1		Bimal-1
Match-2	1	
Match-3		
Match-4	1	Harita-1
Match-5		Bimal-1
Match-6	1	Bimal-1
Match-7		Bimal-1
Match-8	1	Harita-1

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be a each and no. of goals scored in 1st and 5th match be b and c respectively.

Therefore, 2a+b+c= 8

If a=1, then b+c=6 therefore possible solutions for b and c will be 2 and 4 only

If a=2, then b+c=4 therefore possible solution for b and c will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is a=1, b=4 and c=2

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	
Match-3	1	
Match-4	1	Harita-1
Match-5	2	Bimal-1,
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

Matches	Goals Scored	Players
Match-1	4	Bimal-1,Harita-3
Match-2	1	Amla/Sarita-1
Match-3	1	Amla/Sarita-1
Match-4	1	Harita-1
Match-5	2	Bimal-1, Amla/Sarita-1
Match-6	1	Bimal-1
Match-7	1	Bimal-1
Match-8	1	Harita-1

VIDEO SOLUTION

41. **1**

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

	R1	R2	R3	T1	R4	R5	R6	T2
A				6				7
B				-2				-1
C				-2				-5
D	D1	D3	D2	2				-1

**T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

**Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3, D1>D2>D3

D scored 2 points till round R3 and D1>D3>D2 the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

	R1	R2	R3	T1	R4	R5	R6	T2
A	2 H	3 H	1 L	6				7
B	-2 L	-1 L	1 L	-2				-1
C	-2 L	-1 L	1 L	-2				-5
D	2 H	-1 L	1 L	2				-1

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H			7
B	-2 L	-1 L	1 L	-2	-1 L			-1
C	-2 L	-1 L	1 L	-2	-1 L			-5
D	2 H	-1 L	1 L	2	-1 L			-1

For A, R.x+R.y+R.z=1 => R.y+R.z=-2

For B, R.x+R.y+R.z=1 => R.y+R.z=2

For C, R.x+R.y+R.z=-3 => R.y+R.z=-2

For D, R.x+R.y+R.z=-3.=> R.y+R.z=-2

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D: R.y+R.z=-2

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H	-1 L	-1 H	7
B	-2 L	-1 L	1 L	-2	-1 L	3 H	-1 H	-1
C	-2 L	-1 L	1 L	-2	-1 L	-1 L	-1 H	-5
D	2 H	-1 L	1 L	2	-1 L	-1 L	-1 H	-1

Deepak got exactly one point in only R3.

Hence 1 is correct answer.

VIDEO SOLUTION

42. **4**

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

	R1	R2	R3	T1	R4	R5	R6	T2
A				6				7
B				-2				-1
C				-2				-5
D	D1	D3	D2	2				-1

**T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

**Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3, D1>D2>D3

D scored 2 points till round R3 and D1>D3>D2 the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

	R1	R2	R3	T1	R4	R5	R6	T2
A	2 H	3 H	1 L	6				7
B	-2 L	-1 L	1 L	-2				-1
C	-2 L	-1 L	1 L	-2				-5
D	2 H	-1 L	1 L	2				-1

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H			7
B	-2 L	-1 L	1 L	-2	-1 L			-1
C	-2 L	-1 L	1 L	-2	-1 L			-5
D	2 H	-1 L	1 L	2	-1 L			-1

For A, R.x+R.y+R.z=1 => R.y+R.z=-2

For B, R.x+R.y+R.z=1 => R.y+R.z=2

For C, R.x+R.y+R.z=-3 => R.y+R.z=-2

For D, R.x+R.y+R.z=-3.=> R.y+R.z=-2

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D: R.y+R.z=-2

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H	-1 L	-1 H	7
B	-2 L	-1 L	1 L	-2	-1 L	3 H	-1 H	-1
C	-2 L	-1 L	1 L	-2	-1 L	-1 L	-1 H	-5
D	2 H	-1 L	1 L	2	-1 L	-1 L	-1 H	-1

Arun bid high in R1,R2, R.x, R.z hence, 4 is correct answer.

VIDEO SOLUTION

43. **4**

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

	R1	R2	R3	T1	R4	R5	R6	T2
A				6				7
B				-2				-1
C				-2				-5
D	D1	D3	D2	2				-1

**T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

**Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3, D1>D2>D3

D scored 2 points till round R3 and D1>D3>D2 the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

	R1	R2	R3	T1	R4	R5	R6	T2
A	2 H	3 H	1 L	6				7
B	-2 L	-1 L	1 L	-2				-1
C	-2 L	-1 L	1 L	-2				-5
D	2 H	-1 L	1 L	2				-1

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H			7
B	-2 L	-1 L	1 L	-2	-1 L			-1
C	-2 L	-1 L	1 L	-2	-1 L			-5
D	2 H	-1 L	1 L	2	-1 L			-1

For A, R.x+R.y+R.z=1 => R.y+R.z=-2

For B, R.x+R.y+R.z=1 => R.y+R.z=2

For C, R.x+R.y+R.z=-3 => R.y+R.z=-2

For D, R.x+R.y+R.z=-3.=> R.y+R.z=-2

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D: R.y+R.z=-2

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H	-1 L	-1 H	7
B	-2 L	-1 L	1 L	-2	-1 L	3 H	-1 H	-1
C	-2 L	-1 L	1 L	-2	-1 L	-1 L	-1 H	-5
D	2 H	-1 L	1 L	2	-1 L	-1 L	-1 H	-1

Bikram bid Lo in R1,R2,R3,R.x. Hence 4 is correct answer.

VIDEO SOLUTION

44. **2**

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

	R1	R2	R3	T1	R4	R5	R6	T2
A				6				7
B				-2				-1
C				-2				-5
D	D1	D3	D2	2				-1

**T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

**Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3, D1>D2>D3

D scored 2 points till round R3 and D1>D3>D2 the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

	R1	R2	R3	T1	R4	R5	R6	T2
A	2 H	3 H	1 L	6				7
B	-2 L	-1 L	1 L	-2				-1
C	-2 L	-1 L	1 L	-2				-5
D	2 H	-1 L	1 L	2				-1

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H			7
B	-2 L	-1 L	1 L	-2	-1 L			-1
C	-2 L	-1 L	1 L	-2	-1 L			-5
D	2 H	-1 L	1 L	2	-1 L			-1

For A, R.x+R.y+R.z=1 => R.y+R.z=-2

For B, R.x+R.y+R.z=1 => R.y+R.z=2

For C, R.x+R.y+R.z=-3 => R.y+R.z=-2

For D, R.x+R.y+R.z=-3.=> R.y+R.z=-2

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D: R.y+R.z=-2

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H	-1 L	-1 H	7
B	-2 L	-1 L	1 L	-2	-1 L	3 H	-1 H	-1
C	-2 L	-1 L	1 L	-2	-1 L	-1 L	-1 H	-5
D	2 H	-1 L	1 L	2	-1 L	-1 L	-1 H	-1

All the players made identical bids in R3 and R.z

VIDEO SOLUTION

45. **A**

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

	R1	R2	R3	T1	R4	R5	R6	T2
A				6				7
B				-2				-1
C				-2				-5
D	D1	D3	D2	2				-1

**T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

**Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3, D1>D2>D3

D scored 2 points till round R3 and D1>D3>D2 the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

	R1	R2	R3	T1	R4	R5	R6	T2
A	2 H	3 H	1 L	6				7
B	-2 L	-1 L	1 L	-2				-1
C	-2 L	-1 L	1 L	-2				-5
D	2 H	-1 L	1 L	2				-1

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H			7
B	-2 L	-1 L	1 L	-2	-1 L			-1
C	-2 L	-1 L	1 L	-2	-1 L			-5
D	2 H	-1 L	1 L	2	-1 L			-1

For A, R.x+R.y+R.z=1 => R.y+R.z=-2

For B, R.x+R.y+R.z=1 => R.y+R.z=2

For C, R.x+R.y+R.z=-3 => R.y+R.z=-2

For D, R.x+R.y+R.z=-3.=> R.y+R.z=-2

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D: R.y+R.z=-2

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H	-1 L	-1 H	7
B	-2 L	-1 L	1 L	-2	-1 L	3 H	-1 H	-1
C	-2 L	-1 L	1 L	-2	-1 L	-1 L	-1 H	-5
D	2 H	-1 L	1 L	2	-1 L	-1 L	-1 H	-1

The bids by Arun, Bankim, Charu and Dipak, respectively in the first round are HLLH.

Hence Option A is correct.

VIDEO SOLUTION

46. **A**

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

	R1	R2	R3	T1	R4	R5	R6	T2
A				6				7
B				-2				-1
C				-2				-5
D	D1	D3	D2	2				-1

**T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

**Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3, D1>D2>D3

D scored 2 points till round R3 and D1>D3>D2 the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

	R1	R2	R3	T1	R4	R5	R6	T2
A	2 H	3 H	1 L	6				7
B	-2 L	-1 L	1 L	-2				-1
C	-2 L	-1 L	1 L	-2				-5
D	2 H	-1 L	1 L	2				-1

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H			7
B	-2 L	-1 L	1 L	-2	-1 L			-1
C	-2 L	-1 L	1 L	-2	-1 L			-5
D	2 H	-1 L	1 L	2	-1 L			-1

For A, R.x+R.y+R.z=1 => R.y+R.z=-2

For B, R.x+R.y+R.z=1 => R.y+R.z=2

For C, R.x+R.y+R.z=-3 => R.y+R.z=-2

For D, R.x+R.y+R.z=-3.=> R.y+R.z=-2

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D: R.y+R.z=-2

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

	R1	R2	R3	T1	R.x	R y	R.z	T2
A	2 H	3 H	1 L	6	3 H	-1 L	-1 H	7
B	-2 L	-1 L	1 L	-2	-1 L	3 H	-1 H	-1
C	-2 L	-1 L	1 L	-2	-1 L	-1 L	-1 H	-5
D	2 H	-1 L	1 L	2	-1 L	-1 L	-1 H	-1

R2 is correct answer.

VIDEO SOLUTION

47. **B**

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

Group - A	Group - B
Aruna vs	Biju
Azul vs Aruna	Biju vs Brinda/Brij
Azul vs Arif	Biju vs Brinda/ Brij

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

Group - C	Group - D
Chitra vs Dipen	
Chetan vs Chitra	Dipen vs Deb/Donna
Chetan vs Chavvi	Dipen vs Deb/Donna

Aruna plays Chitra in the finals and wins the final round.

Aruna is the winner.

VIDEO SOLUTION

48. **D**

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

Group - A	Group - B
Aruna vs Biju	
Azul vs Aruna	Biju vs Brinda/ Brij
Azul vs Arif	Biju vs Brinda/ Brij

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

Group - C	Group - D
Chitra vs Dipen	
Chetan vs Chitra	Dipen vs Deb/Donna
Chetan vs Chavvi	Dipen vs Deb/Donna

Aruna plays Chitra in the finals and wins the final round.

Chitra and Dipen played in the semifinals

VIDEO SOLUTION

49. **D**

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

Group - A	Group - B
Aruna vs Biju	
Azul vs Aruna	Biju vs Brinda/ Brij
Azul vs Arif	Biju vs Brinda/ Brij

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

Group - C	Group - D
Chitra vs Dipen	
Chetan vs Chitra	Dipen vs Deb/Donna
Chetan vs Chavvi	Dipen vs Deb/Donna

Aruna plays Chitra in the finals and wins the final round.

Brij was the player from group B who played Chitra. Aruna played in finals, Chetan in round 2, and Dipen in semi finals

VIDEO SOLUTION

50. **A**

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

Group - A	Group - B
Aruna vs Biju	
Azul vs Aruna	Biju vs Brinda/ Brij
Azul vs Arif	Biju vs Brinda/ Brij

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

Group - C	Group - D
Chitra vs Dipen	
Chetan vs Chitra	Dipen vs Deb/Donna
Chetan vs Chavvi	Dipen vs Deb/Donna

Aruna plays Chitra in the finals and wins the final round.

Dipen was ranked 2 or 3 in his group

VIDEO SOLUTION

51. **A**

The ranking list would be in the order A, B, C, D......., X, Y, Z. Now the A wins all his 25 matches, B wins 24 matches and lost to A. C wins 23 matches and lost to A and B. In this way N 12 matches and looses 13 matches to A,B,C,D....,M

VIEW SOLUTION

52. **A**

Round no.	Matches	Winners
1	58	58
2	29	29 or (28+1)
3	14	(14+1)
4	7	(7+1 = 8)
5	4	4
6	2	2
7	1	1
Total	115	

In second round, number of winners were odd hence 1 winner will be played in next round. Since in third round also number of winners are odd, then last round player will be played in this round and one another player will be played in next round.

Now in 4th round total winners will be (7+1), this 1 winner belongs to 3rd round where he was sent to play in next round.

Summation of all matches will be 115.

VIEW SOLUTION

53. **A**

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4				
Round 5	10	10	7	13
Round 6				
Round 7		12	4	
Round 8	13			10

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11		
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3		11		10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13	11	6	10

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7	12	12	4	12
Round 8	13	11	6	10

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6	12	11	5	12
Round 7	12	12	4	12
Round 8	13	11	6	10

VIDEO SOLUTION

54. **6**

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4				
Round 5	10	10	7	13
Round 6				
Round 7		12	4	
Round 8	13			10

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11		
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3		11		10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13	11	6	10

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7	12	12	4	12
Round 8	13	11	6	10

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6	12	11	5	12
Round 7	12	12	4	12
Round 8	13	11	6	10

VIDEO SOLUTION

55. **6**

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4				
Round 5	10	10	7	13
Round 6				
Round 7		12	4	
Round 8	13			10

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11		
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3		11		10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13	11	6	10

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7	12	12	4	12
Round 8	13	11	6	10

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6	12	11	5	12
Round 7	12	12	4	12
Round 8	13	11	6	10

The Games which had a bet of Rs.2 are as following
Round 1 - Palak vs Qasim
Round 2 - Qasim vs Suresh
Round 3 - Palak vs Suresh
Round 4 - Ritesh vs Suresh
Round 5 - None
Round 6 - Palak vs Ritesh
Round 7 - None
Round 8 - Ritesh vs Suresh
Hence total number of games that were played with a bet of 2 is 6

VIDEO SOLUTION

56. **D**

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4				
Round 5	10	10	7	13
Round 6				
Round 7		12	4	
Round 8	13			10

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11		
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3		11		10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13	11	6	10

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7	12	12	4	12
Round 8	13	11	6	10

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6	12	11	5	12
Round 7	12	12	4	12
Round 8	13	11	6	10

For round 6 the pairs formed were Pulak-Ritesh and Qasim-Suresh. For round 4 the pairs formed were Pulak-Qasim and Ritesh-Suresh.

Therefore the pairs formed for round 5 were Pulak-Suresh and Qasim-Ritesh

VIDEO SOLUTION

57. **D**

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4				
Round 5	10	10	7	13
Round 6				
Round 7		12	4	
Round 8	13			10

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11		
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3		11		10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13	11	6	10

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7	12	12	4	12
Round 8	13	11	6	10

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6	12	11	5	12
Round 7	12	12	4	12
Round 8	13	11	6	10

VIDEO SOLUTION

58. **D**

From the condition given in the premise, we can make the following table:

	1	2	3	4	5	6	T	7	8	9	10	TT
A		X	X	X	X		8					18
B			X	X	X	X	2					5
C				X	X	X	3	X				6
D					X	X	6	X	X			6
E						X	3	X	X	X		10
F							10	X	X	X	X	10
G	X						17		X	X	X	17
H	X	X					1			X	X	4
I	X	X	X				2				X	17
J	X	X	X	X			14					17

The information known about Rounds 1 through 6:

1. Gordon(G) did not score consecutively in any two rounds.
2. Eric(E) and Fatima(F) both scored in a round.

By observing the table:

1. Jordan(J) scored 7 points in both the rounds 5th & 6th.

2. Amita (A) scored 1,7 points then she scored 7 in the first round.

3. Bala (B) scored 1 point in both the rounds 1st and 2nd.

4. Ikea (I) scored 1 point in the round 4th and 5th.

5. Gordon(G- 7,7,3) did not score consecutively in any two rounds so it scored in 2nd, 4th and 6th rounds respectively.

We can make the following table from the details given in the question.

	1	2	3	4	5	6	T	7	8	9	10	TT
A	7	X	X	X	X	1	8					18
B	1	1	X	X	X	X	2					5
C	3	0	0	X	X	X	3	X				6
D	0	3	0	3	X	X	6	X	X			6
E	0	0	3	0	0	X	3	X	X	X		10
F	0	0	7	0	3	0	10	X	X	X	X	10
G	X	7	0	7	0	3	17		X	X	X	17
H	X	X	1	0	0	0	1			X	X	4
I	X	X	X	1	1	0	2				X	17
J	X	X	X	X	7	7	14					17

T: Total after the sixth round and TT: Total after the 10th round.

1. Only two players scored in three consecutive rounds. One of them was Chen. So He scored 1 point in the rounds 8th, 9th and 10th.

2. Ikea scored 15 points (1,7,7) in three rounds respectively.

3. Eric scored 7 in round 10.

4. Amita will score 3 in round 10, and 7 in round 7.

We can make the following table:

	1	2	3	4	5	6	T	7	8	9	10	TT
A	7	X	X	X	X	1	8	7	0	0	3	18
B	1	1	X	X	X	X	2	0	0	3	0	5
C	3	0	0	X	X	X	3	X	1	1	1	6
D	0	3	0	3	X	X	6	X	X	0	0	6
E	0	0	3	0	0	X	3	X	X	X	7	10
F	0	0	7	0	3	0	10	X	X	X	X	10
G	X	7	0	7	0	3	17	0	X	X	X	17
H	X	X	1	0	0	0	1	0	3	X	X	4
I	X	X	X	1	1	0	2	1	7	7	X	17
J	X	X	X	X	7	7	14	3	0	0	0	17

Hence option D is correct.

VIDEO SOLUTION

59. **D**

From the condition given in the premise, we can make the following table:

	1	2	3	4	5	6	T	7	8	9	10	TT
A		X	X	X	X		8					18
B			X	X	X	X	2					5
C				X	X	X	3	X				6
D					X	X	6	X	X			6
E						X	3	X	X	X		10
F							10	X	X	X	X	10
G	X						17		X	X	X	17
H	X	X					1			X	X	4
I	X	X	X				2				X	17
J	X	X	X	X			14					17

The information known about Rounds 1 through 6:
1. Gordon(G) did not score consecutively in any two rounds.
2. Eric(E) and Fatima(F) both scored in a round.

By observing the table:

1. Jordan(J) scored 7 points in both the rounds 5th & 6th.

2. Amita (A) scored 1,7 points then she scored 7 in the first round.

3. Bala (B) scored 1 point in both the rounds 1st and 2nd.

4. Ikea (I) scored 1 point in the round 4th and 5th.

5. Gordon(G- 7,7,3) did not score consecutively in any two rounds so it scored in 2nd, 4th and 6th rounds respectively.

We can make the following table from the details given in the question.

	1	2	3	4	5	6	T	7	8	9	10	TT
A	7	X	X	X	X	1	8					18
B	1	1	X	X	X	X	2					5
C	3	0	0	X	X	X	3	X				6
D	0	3	0	3	X	X	6	X	X			6
E	0	0	3	0	0	X	3	X	X	X		10
F	0	0	7	0	3	0	10	X	X	X	X	10
G	X	7	0	7	0	3	17		X	X	X	17
H	X	X	1	0	0	0	1			X	X	4
I	X	X	X	1	1	0	2				X	17
J	X	X	X	X	7	7	14					17

T: Total after the sixth round and TT: Total after the 10th round.

1. Only two players scored in three consecutive rounds. One of them was Chen. So He scored 1 point in the rounds 8th, 9th and 10th.

2. Ikea scored 15 points (1,7,7) in three rounds respectively.

3. Eric scored 7 in round 10.

4. Amita will score 3 in round 10, and 7 in round 7.

We can make the following table:

	1	2	3	4	5	6	T	7	8	9	10	TT
A	7	X	X	X	X	1	8	7	0	0	3	18
B	1	1	X	X	X	X	2	0	0	3	0	5
C	3	0	0	X	X	X	3	X	1	1	1	6
D	0	3	0	3	X	X	6	X	X	0	0	6
E	0	0	3	0	0	X	3	X	X	X	7	10
F	0	0	7	0	3	0	10	X	X	X	X	10
G	X	7	0	7	0	3	17	0	X	X	X	17
H	X	X	1	0	0	0	1	0	3	X	X	4
I	X	X	X	1	1	0	2	1	7	7	X	17
J	X	X	X	X	7	7	14	3	0	0	0	17

Hence option D is correct.

VIDEO SOLUTION

60. **B**

From the condition given in the premise, we can make the following table:

	1	2	3	4	5	6	T	7	8	9	10	TT
A		X	X	X	X		8					18
B			X	X	X	X	2					5
C				X	X	X	3	X				6
D					X	X	6	X	X			6
E						X	3	X	X	X		10
F							10	X	X	X	X	10
G	X						17		X	X	X	17
H	X	X					1			X	X	4
I	X	X	X				2				X	17
J	X	X	X	X			14					17

The information known about Rounds 1 through 6:
1. Gordon(G) did not score consecutively in any two rounds.
2. Eric(E) and Fatima(F) both scored in a round.

By observing the table:

1. Jordan(J) scored 7 points in both the rounds 5th & 6th.

2. Amita (A) scored 1,7 points then she scored 7 in the first round.

3. Bala (B) scored 1 point in both the rounds 1st and 2nd.

4. Ikea (I) scored 1 point in the round 4th and 5th.

5. Gordon(G- 7,7,3) did not score consecutively in any two rounds so it scored in 2nd, 4th and 6th rounds respectively.

We can make the following table from the details given in the question.

	1	2	3	4	5	6	T	7	8	9	10	TT
A	7	X	X	X	X	1	8					18
B	1	1	X	X	X	X	2					5
C	3	0	0	X	X	X	3	X				6
D	0	3	0	3	X	X	6	X	X			6
E	0	0	3	0	0	X	3	X	X	X		10
F	0	0	7	0	3	0	10	X	X	X	X	10
G	X	7	0	7	0	3	17		X	X	X	17
H	X	X	1	0	0	0	1			X	X	4
I	X	X	X	1	1	0	2				X	17
J	X	X	X	X	7	7	14					17

T: Total after the sixth round and TT: Total after the 10th round.

1. Only two players scored in three consecutive rounds. One of them was Chen. So He scored 1 point in the rounds 8th, 9th and 10th.

2. Ikea scored 15 points (1,7,7) in three rounds respectively.

3. Eric scored 7 in round 10.

4. Amita will score 3 in round 10, and 7 in round 7.

We can make the following table:

	1	2	3	4	5	6	T	7	8	9	10	TT
A	7	X	X	X	X	1	8	7	0	0	3	18
B	1	1	X	X	X	X	2	0	0	3	0	5
C	3	0	0	X	X	X	3	X	1	1	1	6
D	0	3	0	3	X	X	6	X	X	0	0	6
E	0	0	3	0	0	X	3	X	X	X	7	10
F	0	0	7	0	3	0	10	X	X	X	X	10
G	X	7	0	7	0	3	17	0	X	X	X	17
H	X	X	1	0	0	0	1	0	3	X	X	4
I	X	X	X	1	1	0	2	1	7	7	X	17
J	X	X	X	X	7	7	14	3	0	0	0	17

Hence option B is correct.

VIDEO SOLUTION

61. **D**

From the condition given in the premise, we can make the following table:

	1	2	3	4	5	6	T	7	8	9	10	TT
A		X	X	X	X		8					18
B			X	X	X	X	2					5
C				X	X	X	3	X				6
D					X	X	6	X	X			6
E						X	3	X	X	X		10
F							10	X	X	X	X	10
G	X						17		X	X	X	17
H	X	X					1			X	X	4
I	X	X	X				2				X	17
J	X	X	X	X			14					17

The information known about Rounds 1 through 6:
1. Gordon(G) did not score consecutively in any two rounds.
2. Eric(E) and Fatima(F) both scored in a round.

By observing the table:

1. Jordan(J) scored 7 points in both the rounds 5th & 6th.

2. Amita (A) scored 1,7 points then she scored 7 in the first round.

3. Bala (B) scored 1 point in both the rounds 1st and 2nd.

4. Ikea (I) scored 1 point in the round 4th and 5th.

5. Gordon(G- 7,7,3) did not score consecutively in any two rounds so it scored in 2nd, 4th and 6th rounds respectively.

We can make the following table from the details given in the question.

	1	2	3	4	5	6	T	7	8	9	10	TT
A	7	X	X	X	X	1	8					18
B	1	1	X	X	X	X	2					5
C	3	0	0	X	X	X	3	X				6
D	0	3	0	3	X	X	6	X	X			6
E	0	0	3	0	0	X	3	X	X	X		10
F	0	0	7	0	3	0	10	X	X	X	X	10
G	X	7	0	7	0	3	17		X	X	X	17
H	X	X	1	0	0	0	1			X	X	4
I	X	X	X	1	1	0	2				X	17
J	X	X	X	X	7	7	14					17

T: Total after the sixth round and TT: Total after the 10th round.

1. Only two players scored in three consecutive rounds. One of them was Chen. So He scored 1 point in the rounds 8th, 9th and 10th.

2. Ikea scored 15 points (1,7,7) in three rounds respectively.

3. Eric scored 7 in round 10.

4. Amita will score 3 in round 10, and 7 in round 7.

We can make the following table:

	1	2	3	4	5	6	T	7	8	9	10	TT
A	7	X	X	X	X	1	8	7	0	0	3	18
B	1	1	X	X	X	X	2	0	0	3	0	5
C	3	0	0	X	X	X	3	X	1	1	1	6
D	0	3	0	3	X	X	6	X	X	0	0	6
E	0	0	3	0	0	X	3	X	X	X	7	10
F	0	0	7	0	3	0	10	X	X	X	X	10
G	X	7	0	7	0	3	17	0	X	X	X	17
H	X	X	1	0	0	0	1	0	3	X	X	4
I	X	X	X	1	1	0	2	1	7	7	X	17
J	X	X	X	X	7	7	14	3	0	0	0	17

Hence option D is correct.

VIDEO SOLUTION